Status	Finished
Started	Sunday, 25 October 2026, 8:34 AM
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Duration	7 mins 47 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes the data unsorted
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Dynamic programming
c. Greedy method
d. Backtracking

The correct answer is:
Divide and conquer

Question **3**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data in a single pass
- c. Rearranging data without sorting
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Merging

c. Searching

d. Sorting

The correct answer is: Sorting

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search

b. Merge Sort

c. Binary Search

d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure

b. It divides the list into sublists

c. It recursively sorts subproblems

d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. arrange()
- c. sorted()
- d. order()

The correct answer is: sort()

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are compared element by element
- c. They are split again into smaller sublists
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes code execution slower
- b. Provides a basis for other algorithms to work efficiently
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Selection
- c. Searching
- d. Duplicates

The correct answer is: All of the above

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Quick sort
- c. Merge sort
- d. Selection sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting decreases the efficiency of selection operations
c. Sorting helps in finding duplicates quickly
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It avoids the need for recursion
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data